

THE Q-MODEL — From Q to Cosmology

A Theory of Structure, Complexity and Random Actualization

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This document is released as a *white paper* (research-program description) for archival on Zenodo. It is *not* submitted for peer-reviewed journal publication. The central derivation claim (Einstein recovery from Q-events) remains *logical, not mathematical*: the metric and the BDG action are imported (proven but not TQM-derived), $G = \ell^2 c^3 / \hbar$ is dimensional analysis, and no unique sharp prediction yet discriminates TQM from SM+ Λ CDM. See `Docs/Audits/PublicationReadiness_Final.md`.

Abstract

THE Q-MODEL (TQM) is a theory of structure and content that compresses observable physics to four primitives — the individuation principle Q , Random Actualization, the scale triad $(\ell, \tau, \hbar)$, and a single continuous nonlinearity parameter M^2 — and holds that the *form* of every structure is **derivable** from these primitives while the *content* (specific masses, couplings, multiplicities, the Koide angle) is **realized** by contingent draw. The derivation program yields $U(1)$ as a theorem ($\text{Aut}(S^1) = U(1)$, $\pi_1(S^1) = \mathbb{Z}$; 0.95), spatial dimensionality $d = 3+1$ from the intersection of physical viability windows ($M^2 \approx 5$; 0.85), the multiplicity lower bound $N \geq 3$ from CP violation, the log-normal abundance law from the central limit theorem, and phase-gradient gravity whose leading order is the Einstein equations. The dynamical layer is the graph Laplacian L_Q , whose tight-binding identity $H = tL_Q$ and reversible dynamics yield the Schrödinger equation $i\partial_t\psi = L_Q\psi$ and a Hilbert space of eigenmodes. The non-abelian gauge factors and the Koide relation are classified by a minimal three-category taxonomy (DERIVED / REAL-UNDERIVED / DRAWN) with program consistency 0.95 and zero category conflicts. The theory makes quantitative, falsifiable predictions — the zero-parameter radial acceleration $g_{\dagger} = cH_0/(2\pi)$, a specific $w(z)$, a time-varying $\Lambda(t)$, and the log-normal abundance form — and one such prediction (neutrino-Koide) was already falsified. Five no-go theorems (T-08 through T-12) are conditional statements under the no-new-primitives constraint.

Keywords: THE Q-MODEL; TQM; emergence; quantum foundations; causal set theory; graph Laplacian; Schrödinger equation; gauge structure; radial acceleration relation; dark energy; structure/content split; falsifiability.

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1 Introduction

The Standard Model describes the *content* of particle physics but leaves its *values* unexplained. THE Q-MODEL (TQM) addresses this with a single organizational claim: **structure is derivable; content is realized**.

Every result is classified by a minimal three-category taxonomy:

- **DERIVED** — computable from the primitives by theorem;
- **REAL-UNDERIVED** — real, precise, predictive structure whose origin is not computable (emergent, or structured);
- **DRAWN** — a coincidental draw of the abundance law.

The taxonomy classifies *underivability*; REAL-UNDERIVED and DRAWN are *not* claimed as derivations.

2 Formal Primitive Definitions

2.1 Ontology layer

Primitive	Role
Q	principle of individuation (ontology)
Random Actualization ($\ell, \tau, \hbar$)	genuine ontological chance — assumption A-03 , not derived spacetime scale, clock, action (unit conventions)
M^2	nonlinearity regime (single contingent continuous parameter)

2.2 Dynamical layer (quantum postulates)

1. **Q exists.** Topological charge quanta have position x_i and phase θ_i , and interact pairwise with coupling $J_{ij} = f(|x_i - x_j|)$.
2. **Reversible dynamics.** The state norm $\|\psi\|^2$ is conserved ($\Leftrightarrow$ unitarity).
3. **Born rule.** $P = |\langle \phi | \psi \rangle|^2$, uniquely selected by additivity (Gleason's theorem).
4. **Measurement.** A measurement yields one definite outcome (collapse axiom).

3 Dynamical System Summary

The dynamical content of TQM is the graph Laplacian $L_Q = D - A$ (adjacency A , degree D): real symmetric, positive semi-definite, zero row sums. Its eigenvectors form the *Hilbert space* of the theory. The tight-binding identity (TQM-142) is $(L_Q)_{ij} = 2\delta_{ij} - \delta_{i,j+1} - \delta_{i,j-1}$, giving $H = tL_Q$.

Schrödinger from reversibility (TQM-149–151). For linear evolution $\partial_t \psi = M\psi$, norm conservation forces $M^\dagger = -M$; with the complex structure J ($J^2 = -I$), $M = J \otimes L_Q$ gives

$$i\partial_t \psi = L_Q \psi, \quad \psi(t) = e^{-iL_Q t} \psi(0). \quad (1)$$

Continuum limit (verified). The discrete chain converges to the flat Laplacian with exact spectrum $\lambda_k = (1/\Delta x^2)[2 - 2\cos(\pi k/(N+1))] \rightarrow (\pi k)^2$; the BDG operator converges to the d'Alembertian $\square$ with $O(h^2)$ error. The weighted Laplacian $L_W = D_K - K$ approximates the Laplace–Beltrami operator Δ_g and defines a unitary curved-space Schrödinger equation. (See TQM.Tests/ResearchXC.)

4 Structure/Content Split

Form (topology, symmetry, distribution shape, lower bounds) is **derived**; **content** (drawn values, specific angles, upper bounds) is **contingent**. The split is *falsifiable* (§11): one prediction (neutrino-Koide) was falsified. Two composite objects are admitted: internal $N = 3 = \text{DERIVED (lower bound)} \cap \text{DRAWN (upper bound)}$; $SU(3)$ whole = REAL-UNDERIVED (structure) + DRAWN (count 3).

5 Derivation Hierarchy

Chain	Result
Complexity maximization	$M^2 \approx 5 \Rightarrow d = 3+1$ (0.85)
Oscillation $\rightarrow$ phase $\rightarrow S^1$ vacuum	$U(1)$ (0.95)
Binary defects $\rightarrow Z_2$ winding	$SU(2)$ structure (0.70)
Tri-defect moduli	$SU(3)$ structure (0.10); count DRAWN

6 Complexity Functional

The complexity argument is a **weighted six-component decomposition** of observer viability (ComplexityOptimumAnalyzer.cs), not a single scalar variational functional. $d = 3+1$ is the unique value satisfying Bertrand's theorem, Gauss's law, knot stability, Huygens, and 2 GR polarizations. $M^2 \approx 5$ is the observer-viability intersection. *Honesty note:* a window-intersection argument, not a variational theorem (T-02, 0.85).

7 Gauge Sector

$U(1)$ is DERIVED ($\text{Aut}(S^1) = U(1)$, $\pi_1(S^1) = \mathbb{Z}$; T-01). $SU(2)$ and $SU(3)$ structure are REAL-UNDERIVED (emergent); the color count $n = 3$ is DRAWN. TQM derives the gauge-group *structure*, *not* the gauge *dynamics* (Maxwell / Yang–Mills actions are borrowed, A-07).

8 Flavor Sector

The Yukawa operator is the overlap operator $Y_{ij} = \langle \text{arch}_i | \text{amplitude} | \text{arch}_j \rangle$ (form derived, spectrum underived). The Koide relation $Q = 2/3$ at $\theta = 45^\circ$ is **REAL-UNDERIVED, structured, closed (underivable)** (T-08, 0.70). Neutrino-Koide is **FALSIFIED** (T-11, 0.90): $Q_{\text{max}} = 0.585$ (normal) / 0.500 (inverted) $< 2/3$.

9 Gravity and Emergent GR

$G = \ell^2 c^3 / \hbar$ is a *dimensional* result, not a derivation of gravity's dynamics. The phase-gravity chain (oscillation $\rightarrow$ phase $\rightarrow$ causal density $\rightarrow$ metric $\rightarrow$ curvature $\rightarrow$ Einstein) yields $G_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu} + O(\ell_P^2 R^2)$ at leading order, with two modifications beyond GR: $\Lambda(t) = \alpha / \sqrt{V(t)}$ and singularity resolution at $r \sim \ell_P$.

Honesty note: the chain is **logical, not a new dynamical derivation** — "same equations, same predictions; no falsifiable difference" in the weak field. The value is *ontological*; the physical content beyond GR is confined to $\Lambda(t)$ and singularity resolution. The RAR $g_{\dagger} = cH_0/(2\pi)$ is derived with *zero free parameters* (the 2π is a numerical accident).

10 Cosmology

$\Lambda \sim 1/\sqrt{N}$ is a postdiction; $w(z) = -1 + 0.015(1+z)^{3/2}$; clusters are reproduced; the CMB is an *accepted partial computational layer* (full C_ℓ solver deferred, not a theory gap).

11 Quantitative Predictions

Prediction	Type	Status
RAR $g_{\dagger} = cH_0/(2\pi) \approx 1.05 \times 10^{-10}$	zero-parameter	matches a_0
$w(z) = -1 + 0.015(1+z)^{3/2}$	cosmological	Euclid $> 3\sigma$ by ~ 2030
$\Lambda(t) = \alpha / \sqrt{V(t)}$	dark energy	Euclid / Roman
log-normal abundance law	distribution form	testable
$N \geq 3$ (CP lower bound)	a priori	confirmed
neutrino-Koide $Q = 2/3$	falsifiable	falsified

12 Classification System, No-Gos, Open Questions

The minimal taxonomy classifies fourteen results (consistency 0.95, zero category conflicts, two composites). Five **conditional no-go theorems** (T-08–T-12) bound what the theory cannot do under the no-new-primitives constraint. Every in-scope question is **dispositioned** (derived, underivable, underdetermined, contingent, or accepted as a computational layer); the internal-3 node is dispositioned *unresolved-contingent* (gauge-count no-go T-09 provisional at 0.10).

13 Scope and Limitations

TQM derives the *form* of structure and *classifies* content. It does *not* derive gauge dynamics (A-07), nor contingent values (masses, couplings, multiplicities, Koide angle). Confidence numbers are uncalibrated ordinal ranks. The phase-gravity chain is ontological in the weak field. The unified action is a roadmap, not a result.

14 Conclusion

THE Q-MODEL is *structurally complete* in the precise sense that every in-scope question is **dispositioned**. The single residual of genuine scientific tension is the **internal-3 node** (why internal multiplicity/count saturates at 3), dispositioned unresolved-contingent with a provisional gauge-count no-go (T-09, 0.10). **Structure is derivable. Content is realized.**